

Supplementary Material for SENTRY: Anytime-Valid E-Process Monitoring of LLM Agent Action Streams

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This supplement collects the reproducibility settings, the extended signal definitions, and the Shiryaev–Roberts versus CUSUM discussion referred to in the main paper. Section, equation and reference numbers below are self-contained; equations reused from the main paper are restated here for convenience. All code, the collected trajectory corpus, and the scripts that regenerate every reported number and figure are released with the project repository.

1 REPRODUCIBILITY AND HYPERPARAMETERS

Every number and figure in the paper is regenerated by scripts released with the project. The evaluation is fully deterministic: feature hashing uses a seed-stable CRC rather than Python’s per-process-salted `hash`, so back-to-back runs of the evaluation produce byte-identical output. Randomness enters only through the ten train/calibration/threshold/test splits, which are seeded 0, . . . , 9.

TABLE 1

Hyperparameters. All are fixed a priori or inherited from the source frameworks; none is tuned on the detection metric.

Symbol	Meaning	Value
α	target false-alarm rate	0.2
δ	PAC confidence parameter	0.2
λ_{nov}	novelty weight	4
λ_{IL}	instruction weight	4
$(w_1, \dots, w_4)$	instruction-likeness weights	(3, 2, 2, 1)
κ	power-calibrator exponent	0.5
K	tilt-grid size (exp. mixture)	6
splits	random evaluation splits	10 (seeds 0–9)
min actions	trajectory length filter	2

The reference model is fit only on the training split of *nominal* trajectories; no attack trajectory is seen during fitting or calibration. Data collection used DeepSeek-V4-Flash (served through OpenRouter) as the agent (and, for τ -bench, the simulated-user) backend; the AgentDojo pipeline used the `important_instructions` attack, and the change point of each attack was located by matching the injected string against observation content after alphanumeric normalisation.

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2 EXTENDED SIGNAL DEFINITIONS

Observation instruction-likeness.: Writing o_{t-1} for the tool observation preceding action a_t , the raw instruction-likeness $\text{IL}(o_{t-1})$ is a per-word weighted density of generic instruction cues, and the shipped signal is the excess over a self-calibrating nominal ceiling β ,

$$s_t^{\text{IL}} = \max(0, \text{IL}(o_{t-1}) - \beta), \quad \beta = \max_{o \in \mathcal{D}_{\text{train}}} \text{IL}(o). \quad (1)$$

The lexicon is fixed and generic: imperative verbs (*please, send, ignore, forward, do, must, before, after, execute, run, . . .*), second-person pronouns (*you, your, yours*), pseudo-markup blocks matched by `<[A-Za-z/] [^>]{0,40}>`, and exclamation marks; densities are per word. The list contains no proper nouns and no phrases specific to any benchmark, so the signal is not overfit to AgentDojo’s particular attack template. Because β is the maximum instruction-likeness observed on the training split, the excess in (1) is zero on any observation no more directive than nominal tool output and positive only on genuinely anomalous, instruction-bearing observations.

Argument novelty and tool transition.: The argument-novelty vocabulary is maintained per tool, with a global fallback, over tokens matched by `[A-Za-z0-9_./@-]{2,}`. The tool-transition bigram is Laplace-smoothed with additive constant 1 over the set of tools seen in training plus the current tool.

3 ON THE CHOICE OF SHIRYAEV–ROBERTS VS. CUSUM

SENTRY ships both e-detector recursions. Writing L_n for the per-action baseline increment and M_n for the detector statistic, the Shiryaev–Roberts (SR) and CUSUM recursions are

$$\text{SR: } M_n = L_n(M_{n-1} + 1), \quad M_0 = 0, \quad (2)$$

$$\text{CUSUM: } M_n = L_n \max(M_{n-1}, 1), \quad M_0 = 1. \quad (3)$$

We report SR throughout because, in the mixture-of-increments setting, its additive accumulation ($M_{n-1} + 1$) responds slightly faster to the short attacks in our corpus than CUSUM’s $\max(M_{n-1}, 1)$; the two are known to have matching first-order asymptotic delay under the e-detector framework [Shin, Ramdas and Rinaldo, *New England Journal*

of Statistics in Data Science, 2023], and on our data the difference in detection rate is within the cross-split standard deviation.